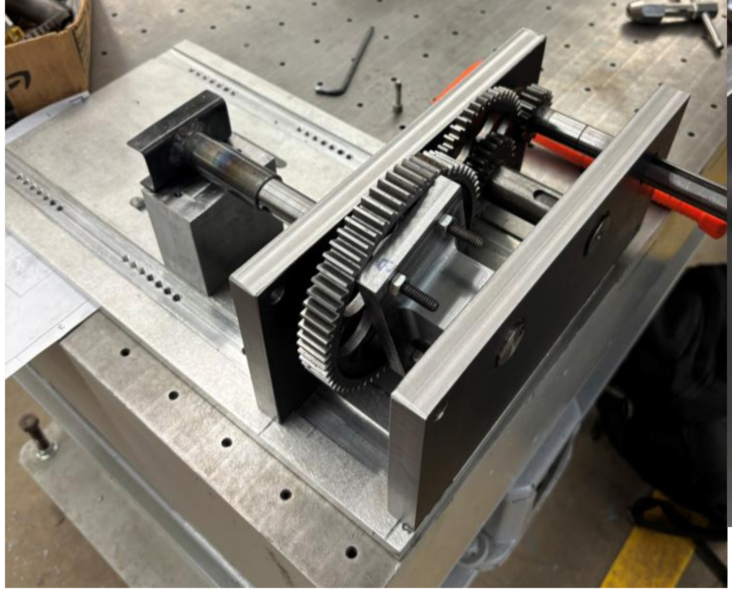
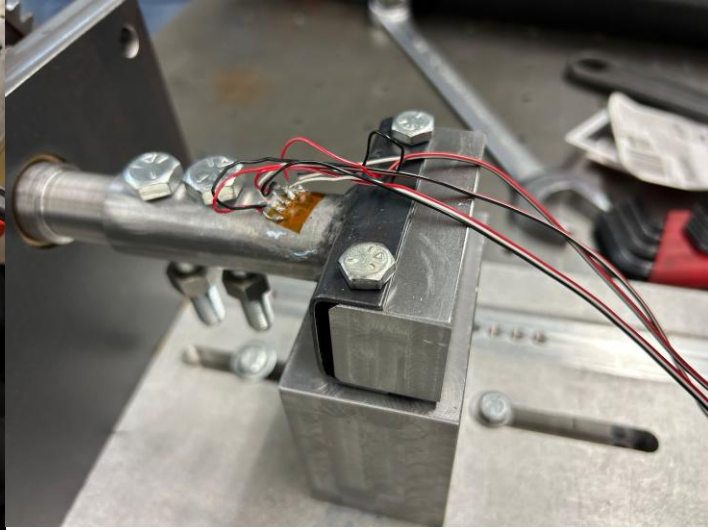
Impact of Heat Treatment on the Yield Strength of Welded 4130 Steel Tube-Flange Assemblies

Goal: Quantify the impact of heat treatment on yield strength of 4130 Steel axle components under torsion. Asses impacts of relieving internal stresses due to welding.

Solution: A custom torque testing rig was created to test annealed, hardened (HRC 30+) and control samples under static torsional loading.





Results: The hardened samples showed a significant increase in yield strength whereas the reduction of internal stresses of the welded flange through annealing did not show a significant improvement.